

# Lecture 8: Profit Maximization under Imperfect Competition



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and Society

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# Full Spectrum of Market Structure

Increasing Competition & Number of Sellers



**Perfect  
Competition**



**Monopolistic  
Competition**



**Oligopoly**



**Monopoly**

Increasing Market Power & Control Over Price

Infinite Sellers  
Homogeneous Product  
No Barriers

Many Sellers  
Differentiated Product  
Low Barriers

Few Large Sellers  
Potential Product Differentiation  
High Barriers

Single Seller  
Unique Product  
Very High Barriers

# Lecture overview

- Monopolistic Competition
- Oligopoly
- Oligopoly and the consumer

# The Spectrum of Imperfect Competition



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- **The Reality of Markets:** Most industries fall between the extremes of Perfect Competition and Pure Monopoly.
- **Imperfect Competition:** Refers to firms that face rivalry yet still possess market power (they are price makers, not price takers).
- **Two Primary Categories:**
  1. Monopolistic Competition: Closer to the competitive end of the spectrum.
  2. Oligopoly: Closer to the monopoly end of the spectrum.

# Monopolistic Competition

# Monopolistic Competition

- A market where many firms compete, but each maintains a monopoly over its specific version of a product.
- Because products are not identical, firms have the power to set prices rather than simply accepting the market rate.

# Monopolistic Competition

- Large Number of Firms: Each firm holds a small market share.
- Freedom of Entry and Exit: New firms can enter the industry easily if they see potential for profit.
- Product Differentiation: products or services are distinct in the eyes of consumers.
  - Market Power: Firms can raise prices without losing all customers because of brand loyalty or unique features.
  - Independence: A firm's actions (like a price cut) won't significantly impact specific rivals, so they don't need to fear immediate retaliation.



# Demand and Examples

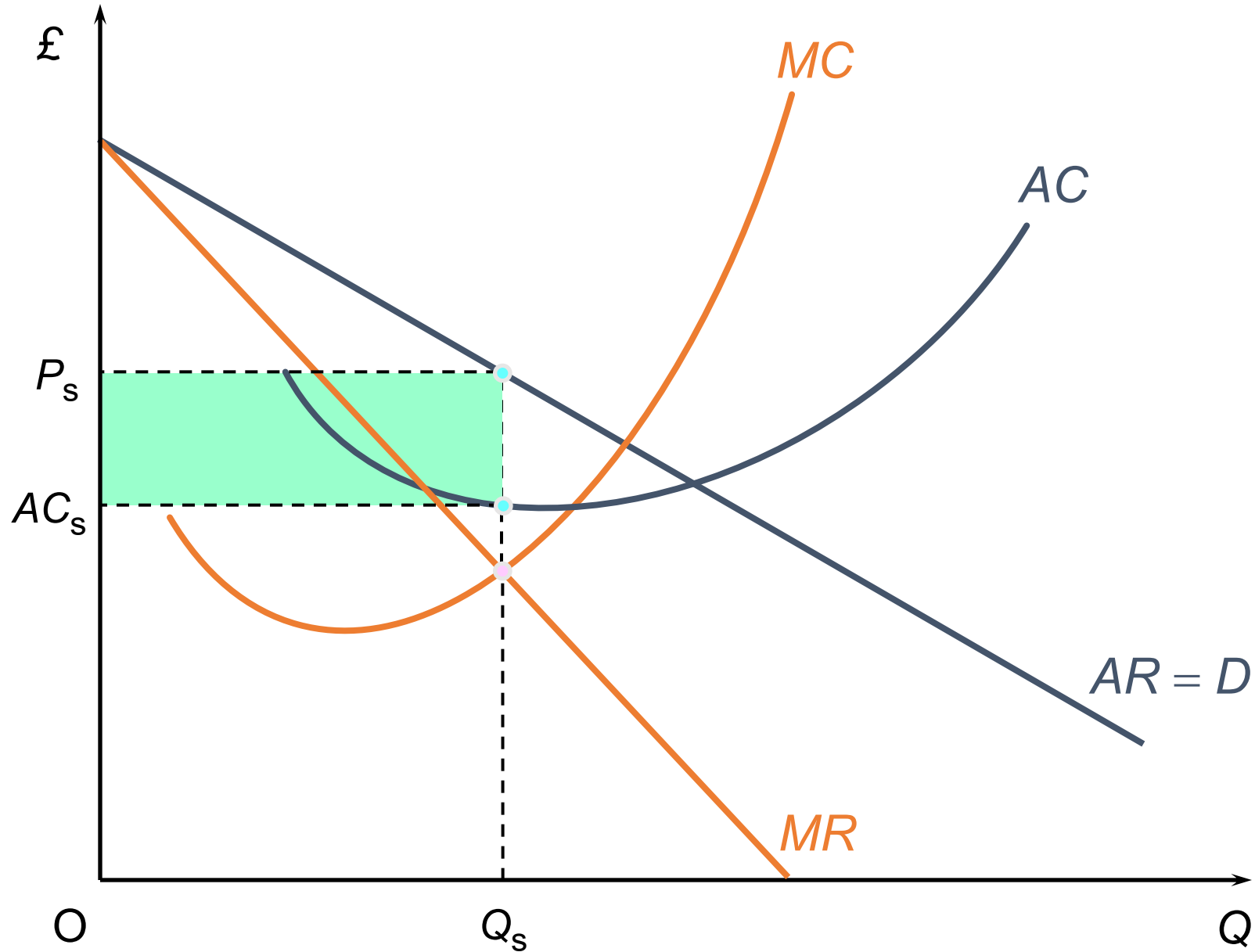
- **Downward Sloping Demand:** Unlike the horizontal curve in perfect competition, the demand curve slopes down but remains relatively elastic due to many available substitutes.
- **Real-World Examples:**
  - Hairdressers and Builders.
  - Restaurants and Petrol Stations.
  - Street performers.



# Monopolistic Competition

- Equilibrium of the firm
  - short run
    - output where  $MC = MR$
    - level of supernormal profit depends on demand
      - position of demand curve
      - elasticity of demand curve

# Short-run equilibrium of the firm under monopolistic competition



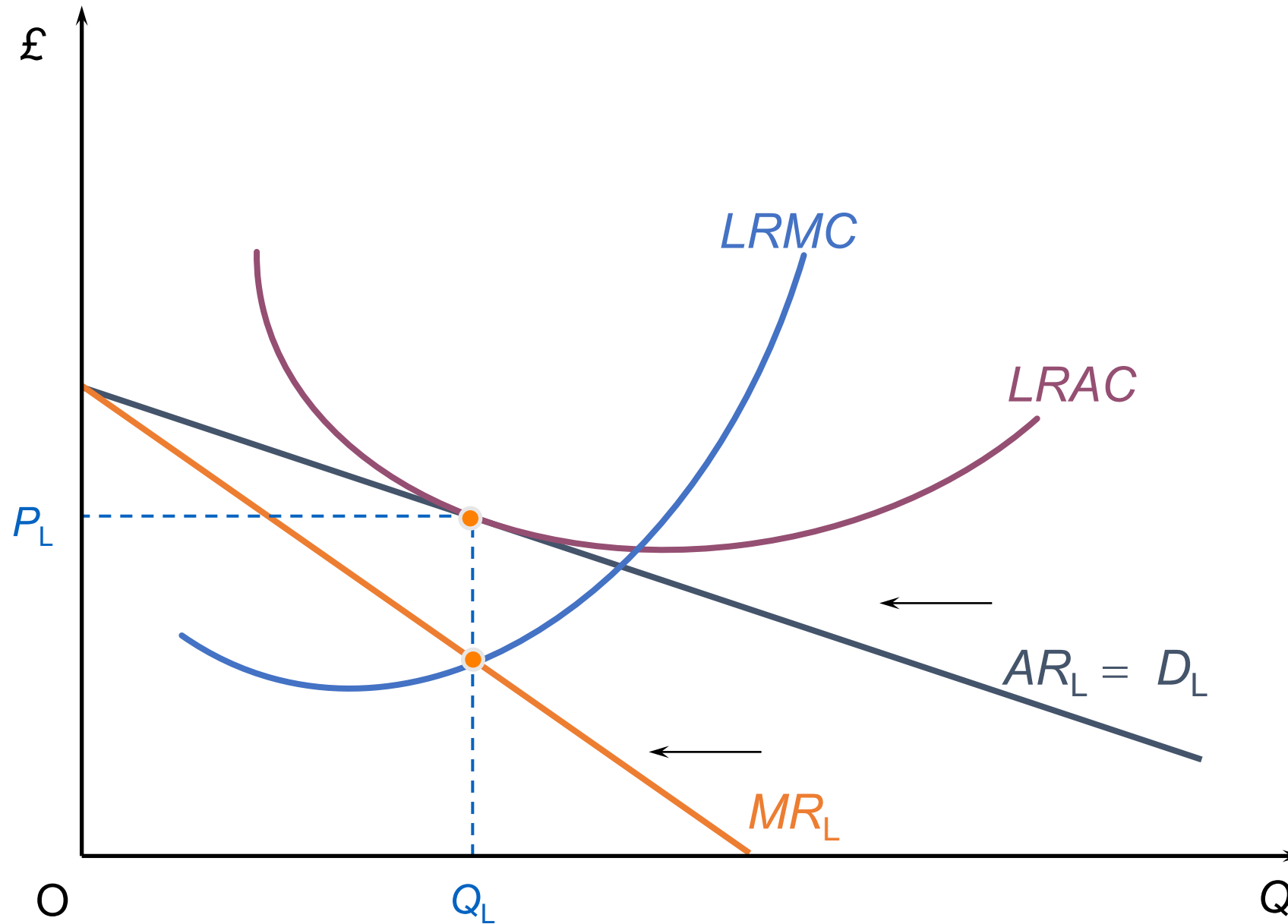
# Monopolistic Competition

- Short-Run Supernormal Profit: Initially, firms earn supernormal profit where Average Revenue (AR) exceeds Average Total Cost (AC) at the profit-maximizing output ( $MR = MC$ ).
- The Signal for Entry: These high profits, combined with low barriers to entry, attract new competitors into the market.
- Increased Competition and Substitutes: As new firms enter, consumers have more varieties to choose from, increasing the number of close substitutes.
- Demand Shift and Elasticity: Each existing firm's demand (AR) curve shifts to the left and becomes more elastic (flatter) because customers can now switch to rivals more easily.

# Monopolistic Competition

- Profit Erosion: This shift reduces the firm's sales and revenue at every price point, shrinking the profit gap.
- Long-Run Normal Profit: Entry continues until  $AR = ATC$  at the new profit-maximizing output, leaving firms with only normal profit.

# Long-run equilibrium of the firm under monopolistic competition



**Q** As new firms enter a monopolistically competitive industry, the demand curve facing each existing firm:

- A.** will shift to the left and become more elastic because there are now more substitutes for its product.
- B.** will shift to the left and become less elastic because there are now more substitutes for its product.
- C.** will not be affected because the new firms do not produce a perfect substitute for its product.
- D.** will shift to the left, but elasticity of demand will not be affected.

# Oligopoly



# Oligopoly

An oligopoly exists when a few dominant firms share the majority of a market's total output.

- **Dominant Players:** Markets are often controlled by household names like Apple, Coca-Cola, Ford, and Tesco.
- **Product Differentiation:**
  - **Identical Products:** Some industries trade in virtually the same goods (e.g., petrol, sugar, chemicals).
  - **Differentiated Products:** Most oligopolies compete through branding and design (e.g., cars, soft drinks, electronics).

## Barriers to Entry

- Unlike monopolistic competition, oligopolies are protected by significant obstacles to entry:
  - Barriers can range from moderate to virtually impossible to overcome.
  - High research and development (R&D) costs, patent protection, and massive advertising budgets.

## Interdependence

The most defining characteristic of an oligopoly is that firms are mutually dependent.

- Every firm must consider how its rivals will react before making a decision.
  - *Example:* If one firm changes its price or launches a new ad campaign, it directly impacts the sales of its competitors.
  - Rivals are likely to respond with their own price changes or promotions.

# Competition vs. Collusion

Interdependence creates a tug-of-war between two conflicting strategies:

## **1. The Incentive to Compete**

- Increase individual market share and maximize a firm's own profit.
- Aggressive price cutting or massive advertising.
- If all firms cut prices, revenue per unit falls; if all increase advertising, costs rise. This often leads to lower profits for everyone.

# Competition vs. Collusion

## 2. The Incentive to Collude

Collusion occurs when firms cooperate to fix prices or output, aiming to reduce market uncertainty and maximize joint profits.

### The Nature of Collusion

- Maximize total industry profits by acting as a single unit. Prevents price wars and expensive retaliatory advertising that erode industry profits.
- Formal or informal agreements on price, output, and product design.
- Firms split the maximized monopoly-style profits.
- Because collusion is harmful to consumers, most countries have competition laws to prevent and punish these agreements.

## Q In oligopoly firms:

- A.

by virtue of their size are able to influence price regardless of whether or not the product is differentiated or standardised.
- B.

are able to influence price only if the products produced by the oligopolists are standardised.
- C.

are able to influence price only if the products produced by oligopolists are differentiated.
- D.

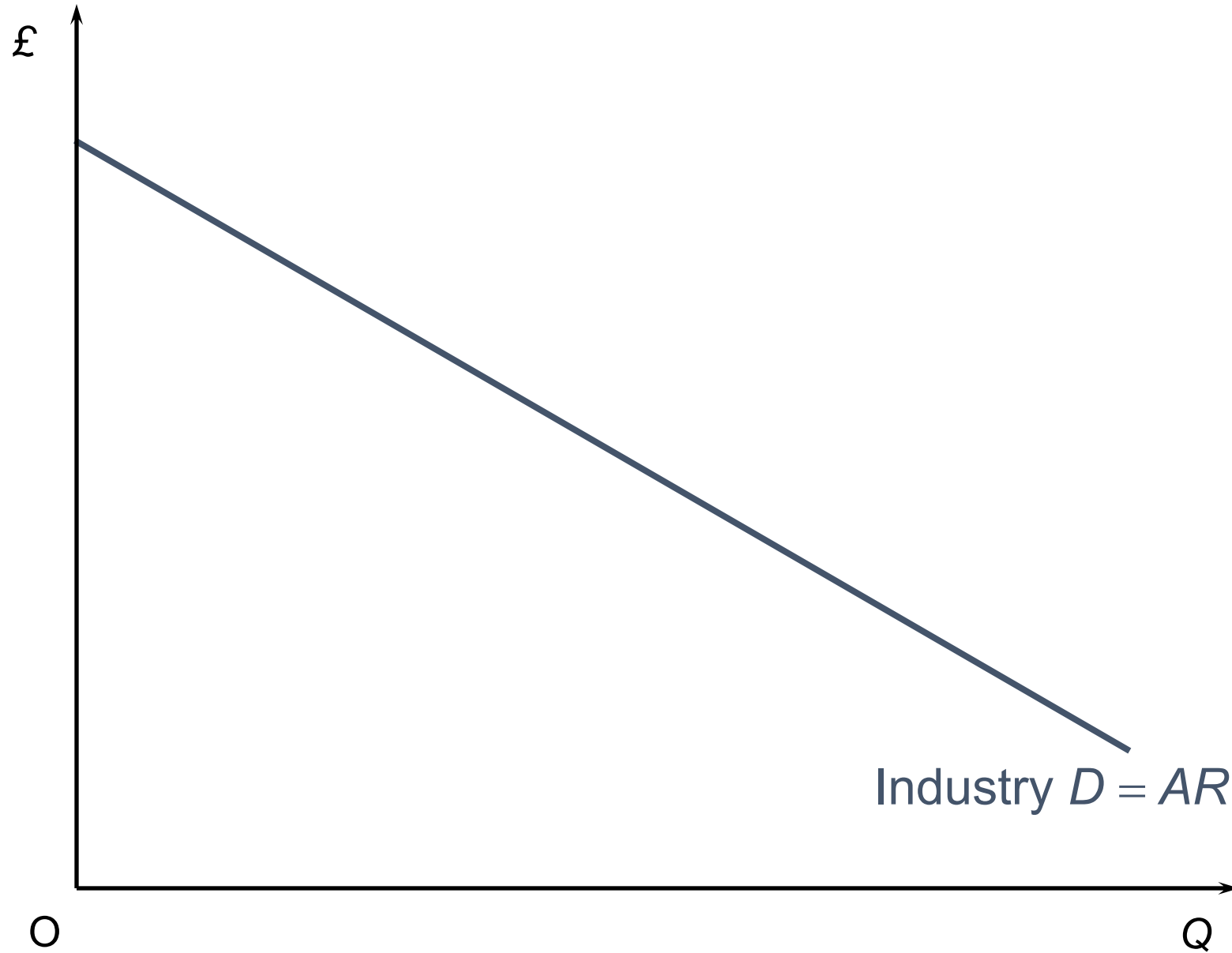
have no influence over price regardless of whether or not the product is differentiated or standardised.

# Collusive Oligopoly & Cartels

The Cartel: A formal agreement where member firms act as a single monopolist.

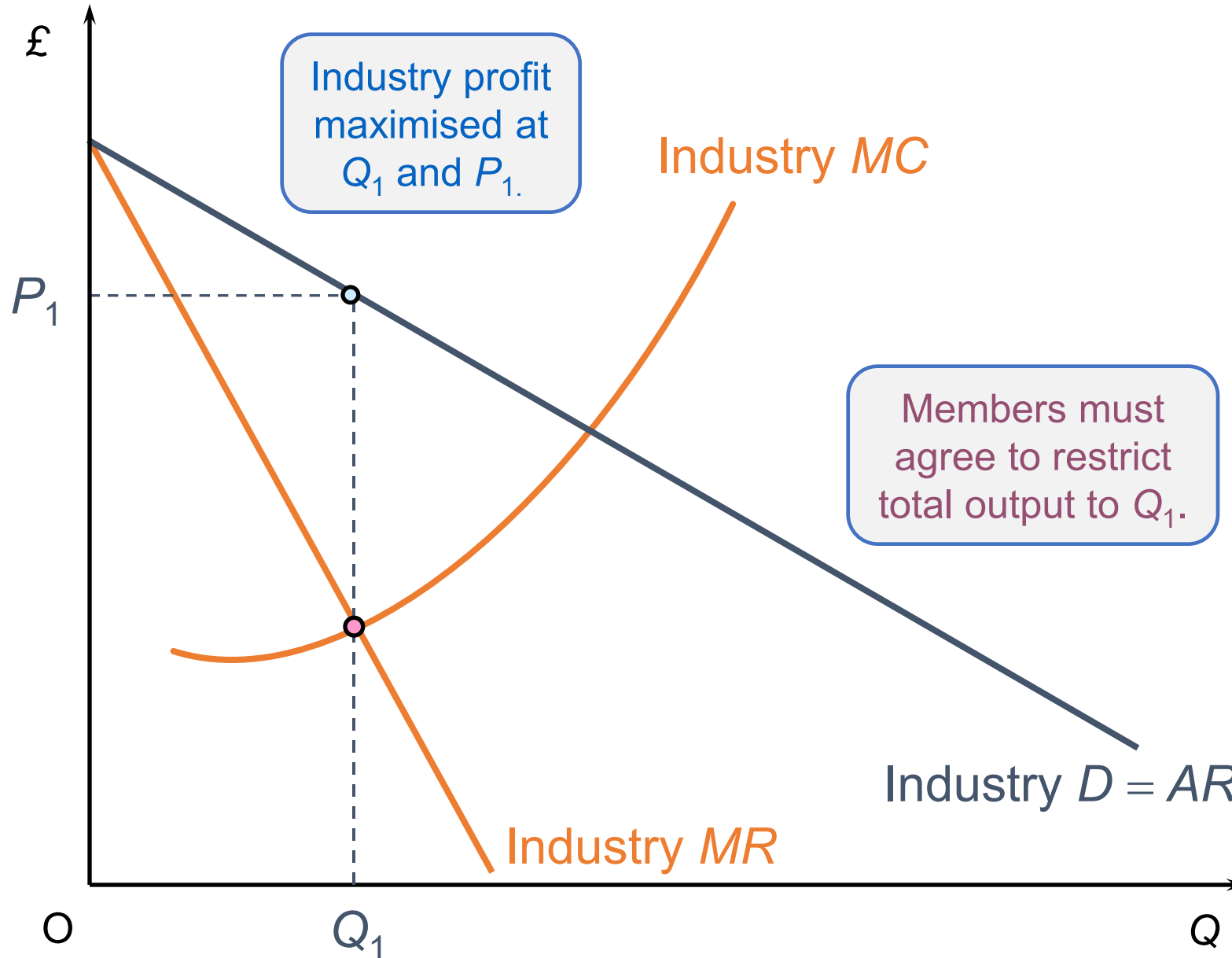
- Profit Maximization:
  - The cartel determines the market price by setting the total output where Market  $MR = \text{Market } MC$ .
  - Market MC is the horizontal sum of all individual members' marginal cost curves.
  - The cartel sets price  $P1$  and total quantity  $Q1$ .

# Profit-maximising cartel





# Profit-maximising cartel



# Collusive Oligopoly & Cartels

## Dividing the Market

Once the price ( $P_1$ ) is set, firms may:

- Engage in Non-Price Competition: Compete for a share of  $Q_1$  through branding and service.
- Implement Quotas: Assign a specific production limit to each member.
  - Quotas must sum exactly to  $Q_1$  to maintain the agreed price.
  - If quotas exceed  $Q_1$ , the market price will collapse due to oversupply.

# Tacit Collusion

Tacit collusion occurs without formal agreements. Firms quietly coordinate behavior by following unwritten rules or market signals.

## Price Leadership

Firms follow the pricing decisions of a lead firm to avoid conflict.

- **Dominant Firm Price Leadership:**
  - A large, powerful firm sets the price first.
  - Smaller follower firms accept this price, assuming the leader has the best market strategy.
- **Barometric Firm Price Leadership:**
  - The leader isn't necessarily the largest, but the most reliable indicator of market shifts.
  - *Example:* When one major bank raises mortgage rates, others quickly follow as the barometer reflects rising costs in the wider economy.

Other Rules of Thumb: Firms may use simplified strategies to stay in sync with rivals:

- Average Cost Pricing: Instead of complex  $MC=MR$  calculations, firms simply add a fixed percentage profit margin to their average costs.
- Price Benchmarks (Price Points): Firms stick to established psychological prices (e.g., £9.99, £24.99).
  - If costs rise, all firms jump to the next benchmark (e.g., £29.99) simultaneously, maintaining price stability across the industry.

# Factors Favouring Collusion

Collusion is most successful when firms trust one another and can easily coordinate their strategies.

## **Ideal Conditions for Collusion:**

- **Small Number of Firms:** It is easier to monitor and reach agreements when only a few well-known rivals exist.
- **Similarity in Production:** When firms have similar costs and production methods, they are more likely to agree on the timing and percentage of price changes.
- **Product Homogeneity:** Reaching a price agreement is simpler if products are similar rather than highly differentiated.
- **Transparency:** Openness regarding costs and production methods builds trust and makes cheating easier to spot.

# Factors Favouring Collusion

- **Market Stability:**
  - Fluctuating demand or costs make agreements difficult to predict and maintain.
  - In a *declining* market, firms are more likely to break agreements and undercut prices to survive.
- **High Barriers to Entry:** Prevents new firms from entering and disrupting the agreement.
- **Lack of Regulation:** No government or legal measures (competition laws) to prevent collusive behavior.

**Q** In which of the following circumstances would a cartel be most likely to work?

- A. The coffee market, where the product is standardised and there are many coffee growers
- B.

 The market for copper, where there are very few producers and the product is standardised
- C. The car industry, where there are few producers but there is great product differentiation
- D. The fast-food market, where there are many producers but the demand for fast food is inelastic

# Oligopoly and the Consumer (The Downsides)

When firms collude, they essentially create a monopoly-lite environment, which can be detrimental to the public.

- **High Prices:** Joint profit maximization leads to prices significantly above marginal cost.
- **Efficiency Issues:**
  - **Scale:** Individual oligopolists may be too small to achieve the same massive economies of scale as a pure monopolist.
  - **Wasteful Spending:** High expenditure on advertising increases average costs, which are then passed on to consumers.



# Potential Benefits of Oligopoly

Despite the risks of collusion, oligopoly can offer unique advantages over other market structures.

- **Countervailing Power:** A saving grace for consumers occurs when oligopolists sell to other powerful buyers (e.g., big supermarkets), who use their own market power to force prices back down.
- **Incentive for R&D:** Unlike a monopolist who faces no rivals, an oligopolist uses supernormal profits for Research and Development to stay ahead of competitors.
  - Improved product design can capture market share before rivals can react.
  - Technological improvements lower costs, helping firms survive potential price wars.
- **Non-Price Competition:** Competition through differentiation leads to greater consumer choice.
  - *Example:* The rapid evolution and variety of smartphones and tablets.

# Questions